ONLINE MEDIA CONSUMPTION DURING COVID-19

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INTRODUCTION

Online media refers to media published over the internet, including web pages, blogs, social media, online newspapers, and video games. Online media has grown remarkably over the years, driven by the digital revolution. The COVID-19 pandemic catalyzed an unprecedented and forced digital migration, transitioning online media from a discretionary utility to the primary medium for social, professional, and educational survival. While this shift ensured operational continuity during global lockdowns, it fundamentally altered the thresholds of daily digital consumption and introduced complex behavioral externalities

This research examines the quantitative shift in media habits during the 2020 lockdown, with a specific focus on the rise in screen time and its corresponding psychological impact on consumers. By analyzing primary survey data, this study moves beyond descriptive observation to test the statistical significance of these behavioral changes.

SIGNIFICANCE OF THE STUDY:

This topic is very important in the current scenario, as the lack of social interaction, various restrictions due to the pandemic, and the virtual continuation of day-to-day proceedings have caused people to become even more dependent on online media than before. The understanding of these shifts is not merely an academic exercise; it is critical for developing data-driven crisis management strategies. As digital integration becomes permanent, identifying the point where utility becomes a psychological burden is essential for policymakers and mental health professionals to assist with mitigating the long-term effects of "digital burnout".

RESEARCH OBJECTIVES: -

1. To quantify the shift in daily hours spent on digital platforms compared to pre-pandemic numbers

2. To identify the correlation between high-volume usage and self-reported mental health indicators, such as anxiety and social discomfort.

3. To assess the proliferation of misinformation and the vulnerability of consumers to "fake news" during periods of heightened digital reliance

DATA COLLECTION AND METHODOLOGY

This study utilized primary data collection through a structured digital survey distributed between 2021 and 2022. The survey was designed to capture self-reported changes in online media usage across two distinct temporal windows: the pre-pandemic baseline and the active lockdown period.

- **Sample Size** Cohort of 40 participants
- **Number of Variables:** There are 16 variables equivalent to the number of questions asked in the questionnaire
- **Survey Instrument:** Digital questionnaire consisting of categorical usage metrics, Likert-scale psychological assessments, and open-ended qualitative responses. The questionnaire was created on Google Forms and was passed to the respondents through social media platforms.

  The link to the Google Form is: - https://forms.gle/BFbW2vvbWvT1tihG9
- **Integrity Measures:** Survey responses were anonymized to encourage honest reporting on sensitive topics like mental health and social discomfort
- **Sample Technique:** Convenience sampling was used to collect the data
- **Sampling Bias:** Due to convenience sampling, there is a possibility of sampling bias. The potential for bias lies in the fact that the form was mainly sent to respondents in the age group of 18-30.

ANALYTICAL FRAMEWORK

1. **Data Preprocessing and Descriptive Statistics:**
   Raw survey data were audited for completeness. The variables were categorized demographically, and the basic statistical measures were calculated for each variable to get a sense of the data.

2. **Hypothesis Testing:**
   To determine if the observed increase in media usage was statistically significant, a Two-Proportion Z-test was performed. This tested the null hypothesis (H0) that the proportion of high-frequency users (3+ hours) remained constant across both periods. The test statistic is: -

$$Z = \frac{p1 - p2}{\sqrt{PQ\left(\frac{1}{n1} + \frac{1}{n2}\right)}}$$

3. **Relational Analysis:**
   The study examined the relationship between the perceived "helpfulness" of media and its "mental health impact". By utilizing Correlation Matrices, the study identifies if high consumption of "Education/Work" acted as a buffer or a contributor to reported anxiety and social discomfort

TECHNOLOGY STACK:

1. **Data Acquisition: Google Forms** was utilized to collect the data, and **Microsoft Excel** was used to get the data in an organized form.

2. **Data Engineering: Python (Pandas)** was used for pre-processing, specifically for cleaning demographic cohorts and for converting multi-response survey strings into binary indicators for analysis.

3. **Statistical Computing: Python (SciPy)** was utilized to execute inferential tests, providing high-precision validation of behavioural shifts.

4. **Data Visualization: Python (Seaborn)** was used to generate comparative distributions and correlation heatmaps.

ANALYSIS / RESULTS

OBJECTIVE 1: Analysis of Consumption Pattern

TABLE NO: 1

Hours Spent Before Covid-19	Frequency	Percentage
0-1	6	15%
1-2	17	42.5%
2-3	10	25%
3+	7	17.5%

TABLE NO: 2

Hours Spent During Covid-19	Frequency	Percentage
0-1	0	0%
1-2	4	10%
2-3	12	30%
3-4	24	60%

The data in Table No. 1 and Table No. 2 confirm an unprecedented "digital migration" during the lockdown. Before the pandemic, only **17.5%** of the cohort reported usage exceeding 3 hours. During the pandemic, this "high-frequency" tier became the majority, with **60%** of respondents consuming 3–4 hours of online media daily.

Statistical Significance: A **Two-Proportion Z-test** was employed to determine if this increase was a statistically significant shift.

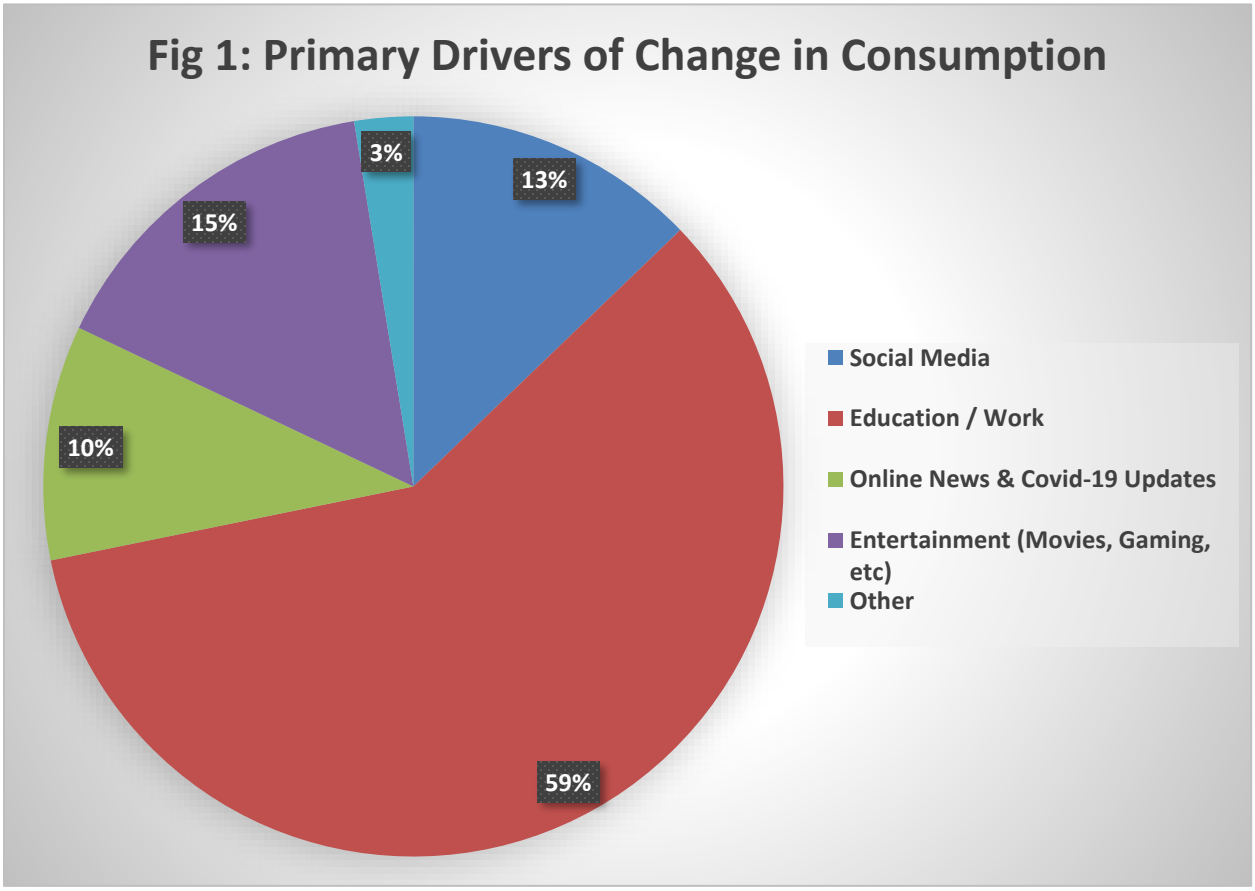
H0: There is no significant difference between the proportion of people consuming online media for 3+ hours pre-COVID and post-COVID years

H1: The proportion of people consuming online media for 3+ hours is significantly higher in the post-COVID years compared to the pre-COVID years

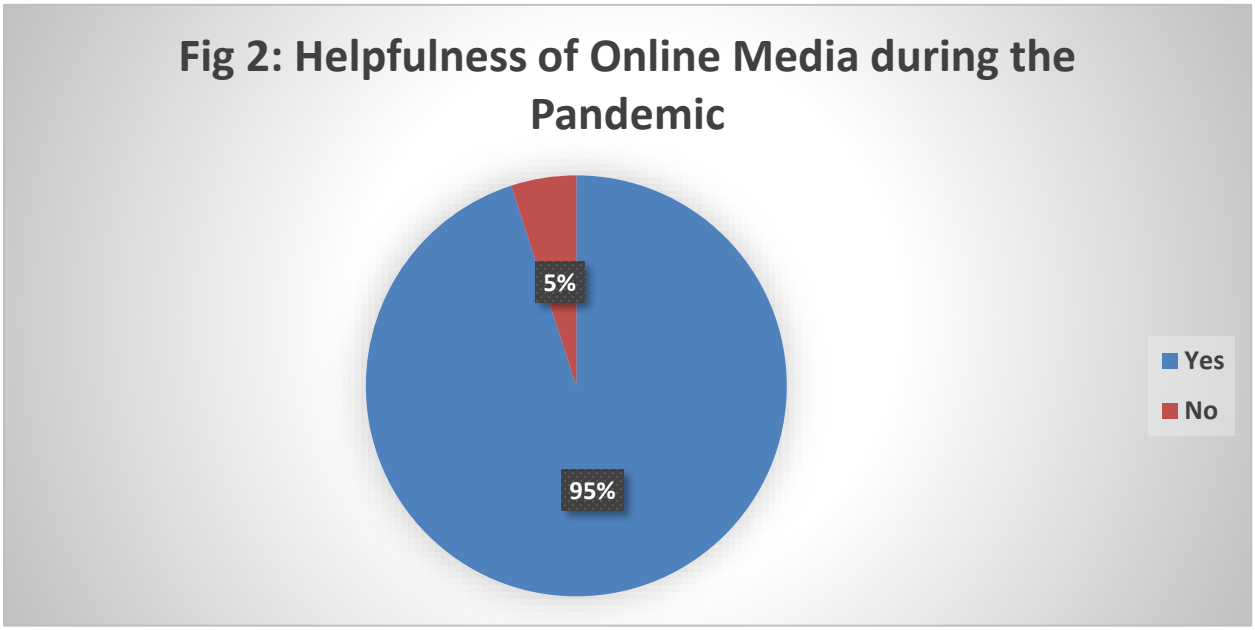
Significance Level = 0.05 or 5%

Test Result: The p-value is 0.00005, which is less than the significance level of 0.05.

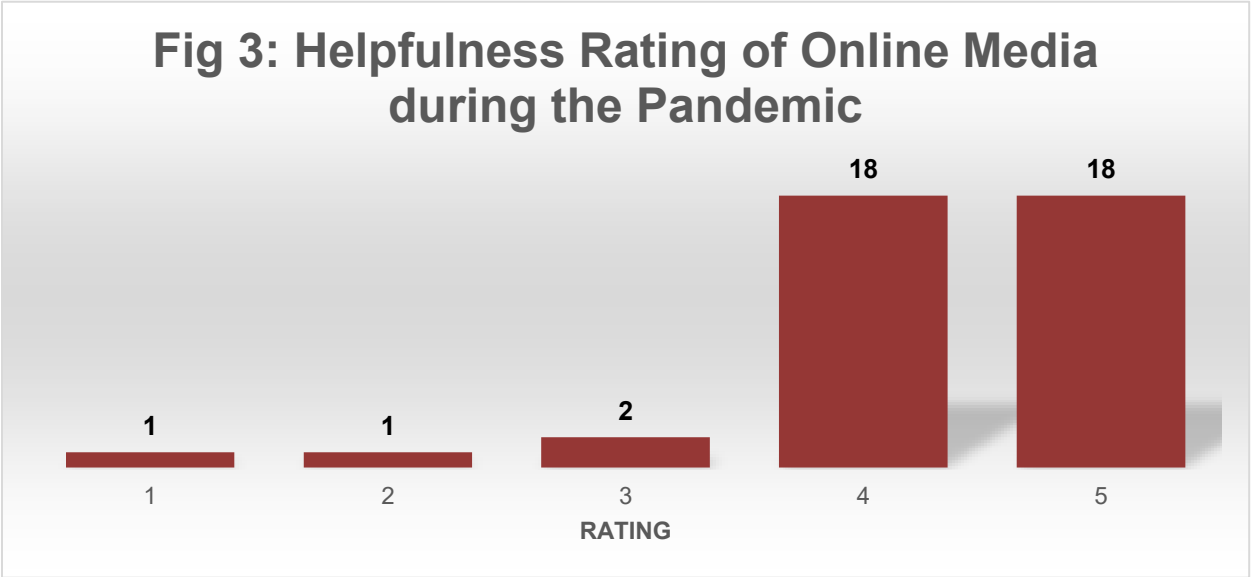
Thus, we reject the H0, confirming that the pandemic caused a statistically significant increase in daily digital reliance. From Figure 1, it is evident that this change was primarily driven by the forced transition of professional and educational activities to online modes



OBJECTIVE 2: Perceived Utility Vs Psychological Cost

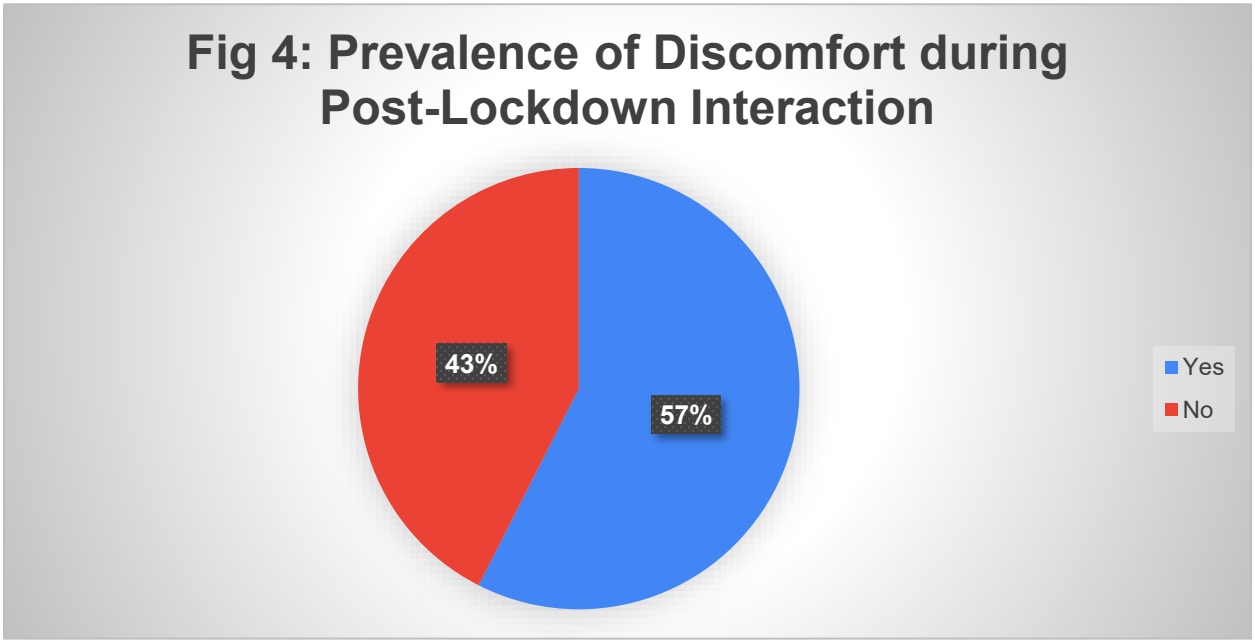


From Figure 2, **95%** of respondents categorized online media as "helpful," largely due to its role in curing boredom and facilitating remote work/education

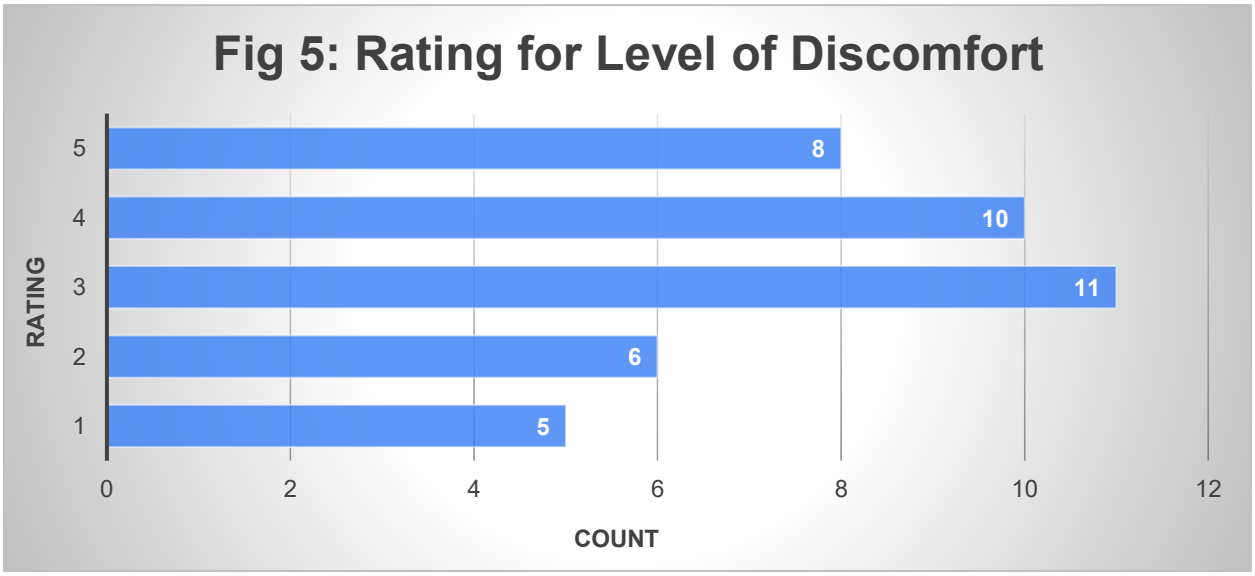


The helpfulness rating of online media ranges from 1(Not helpful at all) to 5(Extremely helpful). From Figure 3, the majority feel that online media was extremely helpful, as most of the ratings are 4 or 5.

The mean helpfulness rating reached **4.3/5.0**, indicating that online media served as a vital "survival utility" during global lockdowns.



From Figure 4, it is clear that people did face some discomfort interacting with people after lockdown, as 57% of them voted 'Yes'.



The discomfort rating of face-to-face interaction ranges from 1(Very uncomfortable) to 5 (Extremely comfortable).

Post-lockdown comfort levels for face-to-face interaction yielded a mean score of **3.25/5.0.**

Statistical Significance: A **Two-Proportion Z-test** was employed to compare the high comfort (4-5) and low comfort zones (1-2)

H0: There is no significant difference between the proportion of people who rated 1 and 2 on comfort level and the proportion of people who rated 4 and 5

H1: There is a significant difference between the proportion of people who rated 1 and 2 on comfort level and the proportion of people who rated 4 and 5

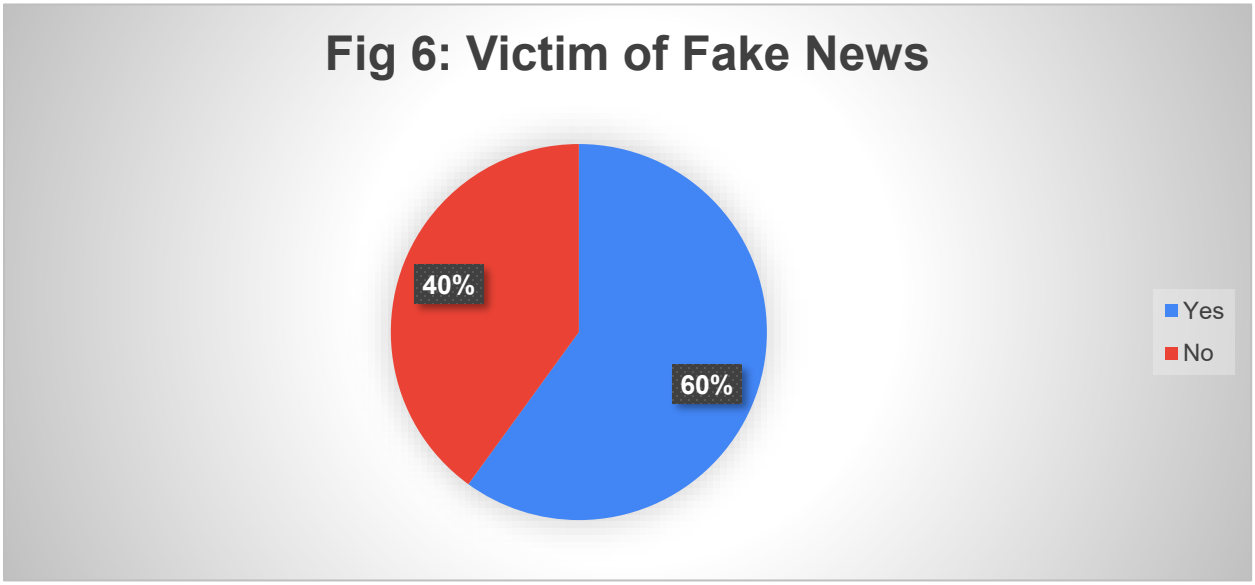
Significance level = 0.05 or 5%

Test Result: The p-value is 0.104, which is more than the significance level of 0.05.

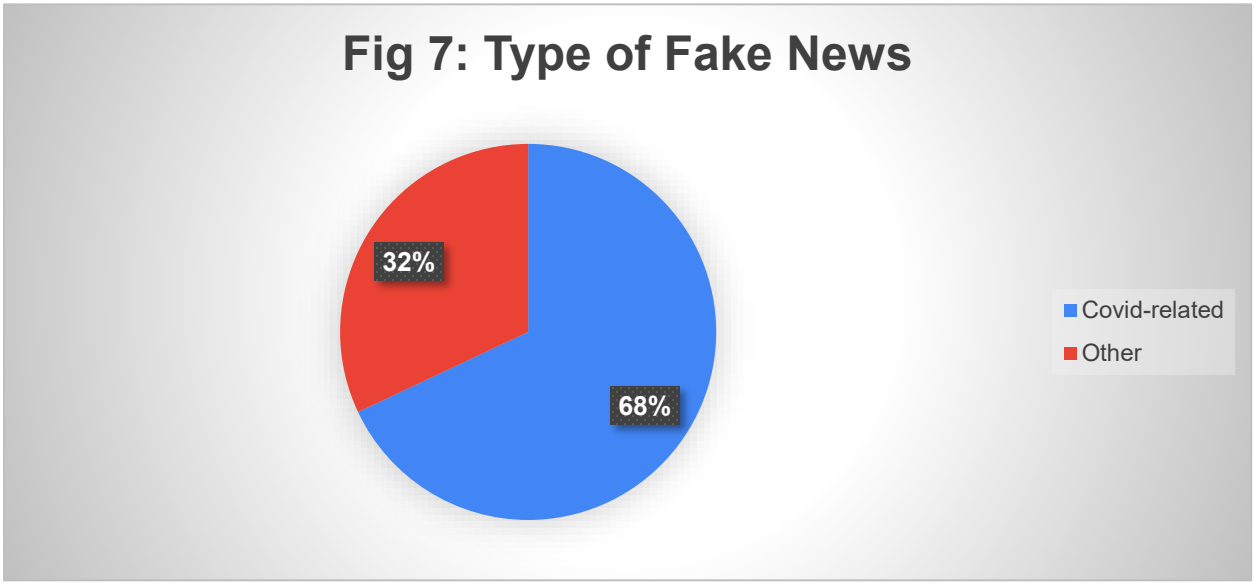
Thus, we accept H0. This suggests that while there isn't a dominant "discomfort", the population remains in a state of "social neutral," neither fully comfortable nor uncomfortable with physical re-entry.

OBJECTIVE 3: Information Integrity and Misinformation Vulnerability

As digital platforms became the primary information source, the study noted a significant rise in "misinformation" vulnerability.



Approximately **60%** of respondents admitted to unknowingly believing misinformation during the pandemic



Data indicates that **COVID-related misinformation,** ranging from natural herb "cures" to virus origin conspiracies, was more likely to be believed than other news categories.

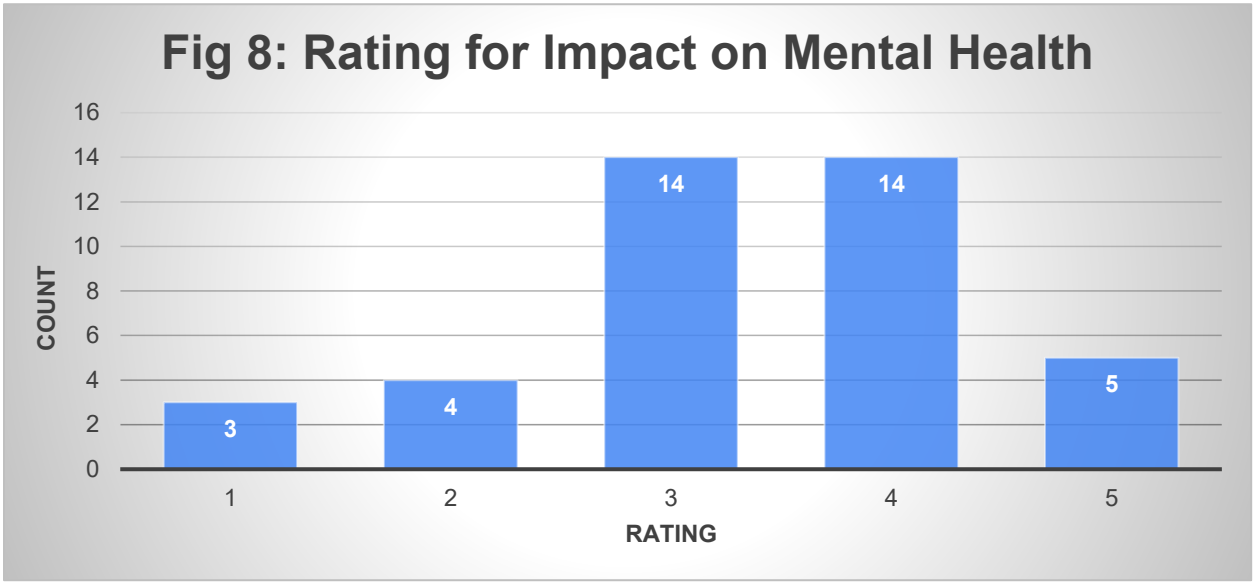
Heightened digital reliance during a crisis creates a "vulnerability window" where the speed of information consumption outpaces the consumer's ability to verify facts, leading to widespread exploitation. A primary reason for the exploitation was more engagement from social media users,

OBJECTIVE 4: Impact on Mental Health

TABLE NO: 3	
Answer	**Count**
Yes	23
No	17

The above table shows the answers to the question whether online media had an impact on the respondents mental health during the pandemic.

While **57.5%** of the cohort (23 out of 40 reported) confirmed a direct impact on their mental health, the severity of this impact was the focal point of testing.



From the above figure, the mental health of most people was atleast partially impacted by online media during the pandemic.

The mean score of impact on mental health was **3.35** provides further evidence to the inference obtained from the graph that people were atleast partially impacted by online media during the pandemic.

Statistical Significance: A **Z-test** was employed to compare the high-impact ratings (4–5) against low-impact ratings (1–2).

H0: There is no significant difference between the proportion of people who gave ratings 4 and 5 and the proportion of people who gave ratings 1 and 2.

H1: The proportion of people who gave ratings 4 and 5 is significantly greater than the proportion of people who gave ratings 1 and 2

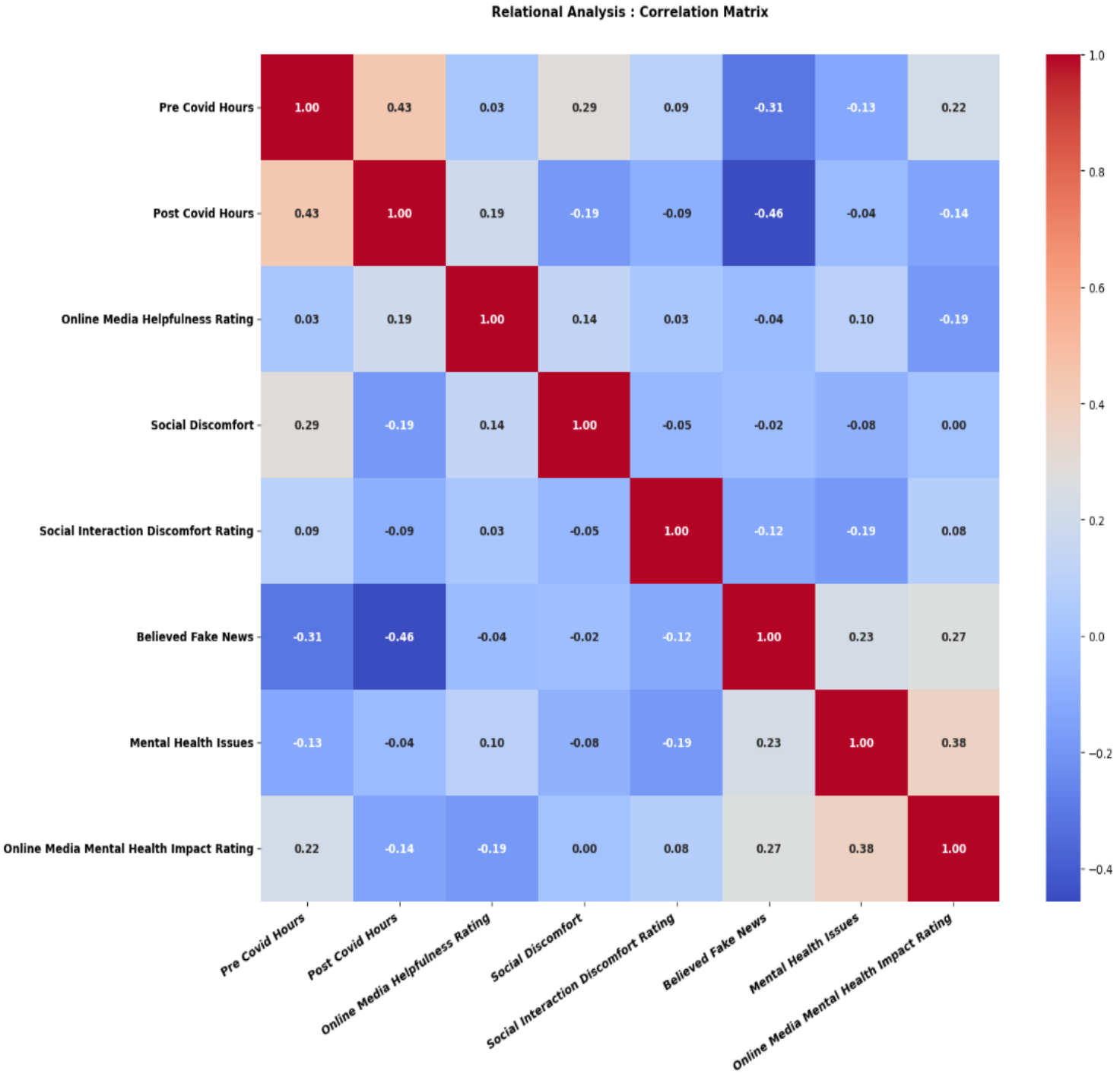
Significance level = 0.05 or 5%

Test Result: The p-value is 0.0021, which is less than the significance level of 0.05

Thus, we reject H0. So the proportion of respondents suffering significant mental health impacts (anxiety, digital burnout) is significantly greater than those reporting minimal impact.

OBJECTIVE 5: Relational Analysis

While the previous sections analyzed variables in isolation, this section examines the interdependence between digital habits and psychological outcomes. By utilizing a Pearson Correlation Matrix, we can identify the strength and direction of these relationships.



Based on the statistical processing of the cohort, several significant patterns emerged:

- **The Mental Health Severity (r = 0.38):** There is a moderate positive correlation between the binary report of 'Mental Health Issues' and the 'Mental Health Impact Rating' (1-5). This validates that as individuals identified the presence of an issue, the perceived severity increased proportionately.
- **The Misinformation-Stress Connection (r = 0.27):** A positive correlation exists between 'Believing Fake News' and the severity of 'Mental Health Impact'. This supports the hypothesis that vulnerability to misinformation acted as a psychological stressor during the height of the pandemic.
- **The Consumption Paradox (r = -0.46):** Counter-intuitively, 'Post-Covid Hours' showed a moderate negative correlation with 'Believing Fake News'. This suggests that for this cohort, higher frequency of use did not lead to higher vulnerability, likely because the usage was driven by **Education/Work** rather than aimless scrolling.
- **The Social Bridge Effect (r = -0.19):** High 'Post-Covid Hours' were negatively correlated with 'Social Discomfort'. This implies that for many respondents, the digital medium served as a necessary social lifeline that mitigated the discomfort of isolation, rather than exacerbating the friction of physical re-entry.

CONCLUSION

The COVID-19 pandemic was not merely a public health crisis; it was a forced, global experiment in digital saturation. This research confirms that the "forced digital migration" in 2020 was not a temporary pivot, but a permanent structural shift in human behaviour. While the study has a limited cohort, the sample size was large enough to produce statistically significant results.

The mental health impacts confirm that our reliance on digital platforms has reached a critical, systemic threshold. We have built a world where "high frequency" usage is no longer discretionary but mandatory for professional and educational survival. However, the math proves that this infrastructure is currently built on a foundation of anxiety.

The "Digital Paradox" identified in this study reveals that while online media achieved a high utility rating (4.3/5.0), it failed to foster true connectivity. The data on social discomfort (Mean Score: 3.25/5.0) suggests that extended virtual interaction acts as a poor substitute for physical presence, leading to a state of "social neutral" where individuals are neither fully isolated nor authentically connected. We are more reached than ever, yet we report increased introversion and social friction. At the same time, the higher online media usage correlated with lower social discomfort, indicating that online media mitigated the friction of physical re-entry.

Perhaps the most alarming finding is the 60% vulnerability rate to misinformation. In our haste to seek information during a crisis, we effectively traded our critical discernment for speed. But it was seen that increased online hours actually correlated with decreased vulnerability to fake news. This suggests that the "forced migration" for Education and Work may have inadvertently trained users to be more discerning or "platform aware."

Moving forward, the challenge for developers, policymakers, and consumers is to move beyond mere connectivity and toward Digital Resilience. We must ask: At what point does a tool for survival become a weight on the human psyche? The data suggests we have already passed that point; the next phase is not about more access, but about reclaiming the boundaries of our digital lives to mitigate the long-term effects of chronic digital burnout.